

Downloading and Compiling SP_Ace / SPACE

SP_Ace is the Stellar Parameters and Chemical Abundances Estimator. It is a Fortran95 spectral-synthesis code that compares an observed stellar spectrum with models generated from the GCOG library. The code estimates stellar parameters such as T_{eff} , $\log g$, $[M/H]$, $[Fe/H]$, and, when the data are good enough, abundance information for other elements or for grouped alpha/metals fits.

1. Install the required compiler tools

On Linux or Windows Subsystem for Linux, open a terminal and run:

```
sudo apt update
sudo apt install build-essential gfortran make unzip dos2unix
```

On macOS, install command-line tools and gfortran through Homebrew:

```
xcode-select --install
brew install gcc make
```

On Windows, the recommended approach is to use Ubuntu through WSL. Do not try to compile SP_Ace in a normal Windows command prompt unless you already know how to build Fortran programs there.

2. Download SP_Ace and the GCOG library

Go to the official GAVO SP_ACE page and follow the local-download link. Download both:

```
SP_Ace / SPACE source-code package
GCOG library package
```

Keep the two downloads together. The GCOG library is not optional; SP_Ace cannot run without it.

For this NGC 6569 project, be careful not to replace the project's working library blindly. Our AAT analysis used the local library directory named:

```
GCOGLibrarySPACE1/
```

This library must support the red CaT-region wavelength windows used in our project. If a newly downloaded GCOG library only supports a different wavelength range, it will not reproduce our AAT CaT-region results.

3. Recommended directory structure

Create a working directory like this:

```
mkdir -p ~/SPACE  
cd ~/SPACE
```

After unzipping the downloaded files, the directory should look something like this:

```
SPACE/  
  SP_Ace_source/  
    Makefile  
    *.f  
    *.f90  
    other source files  
  
  GCOGLibrarySPACE1/  
    library files used by SP_Ace  
  
  spectra/  
    aat_215.txt  
    aat_069.txt  
    other rest-frame spectra
```

The exact source directory name may differ depending on the downloaded package.

4. Compile SP_Ace

Move into the SP_Ace source-code directory:

```
cd ~/SPACE/SP_Ace_source
```

Compile:

```
make
```

If compilation succeeds, an executable named something like this should appear:

```
SPACE
```

Check that it exists:

```
ls -lh SPACE
```

Make sure it is executable:

```
chmod +x SPACE
```

Copy it into the main working directory:

```
cp SPACE ~/SPACE/
```

Then return to the working directory:

```
cd ~/SPACE
```

5. If compilation fails

First check whether `gfortran` is installed:

```
gfortran --version
```

If the `Makefile` expects a different compiler, open it and look for a line such as:

```
FC = ifort
```

or

```
F90 = ifort
```

Change it to:

```
FC = gfortran
```

or:

```
F90 = gfortran
```

Then try again:

```
make clean
make
```

If Windows line endings cause problems, run:

```
dos2unix Makefile
dos2unix *.f
dos2unix *.f90
make clean
make
```

6. Test that SPACE can run

From the working directory, check that the executable responds:

```
./SPACE
```

If there is no `space.par` file yet, it may stop with an error. That is fine. It means the executable exists and is being called.

7. Create a test `space.par`

For one rest-frame AAT spectrum, create a parameter file named:

```
space.par
```

A typical NGC 6569 conservative setup is:

```
obs_sp_file 'spectra/aat_215.txt'
GCOGlib 'GCOGLibrarySPACE1/'
fwhm 0.8

wave_lims 8450 8493 8503 8535 8550 8660 8670 8745

no_norm
error_est
null_value 'NaN'
```

```
ABD_loop  
Salaris_MH
```

The important settings are:

```
obs_sp_file
```

This is the spectrum to fit. It should be an ASCII file with wavelength in Angstroms and normalized flux.

```
GCOGlib
```

This points to the GCOG library directory.

```
fwhm
```

This is the starting estimate of the instrumental FWHM in Angstroms. For the AAT CaT spectra, we used approximately 0.8 Å.

```
wave_lims
```

These are the wavelength windows used for the fit. For the conservative NGC 6569 run, we used:

```
8450-8493 Å  
8503-8535 Å  
8550-8660 Å  
8670-8745 Å
```

These windows avoid the CaT line cores and avoid the worst sky-line region at longer wavelengths.

```
no_norm
```

This tells SP_Ace not to renormalize the spectrum internally. We use this only because our spectra have already been continuum normalized and manually inspected.

```
error_est
```

This asks SP_Ace to estimate uncertainties. It makes the run slower.

```
ABD_loop  
Salaris_MH
```

These are used for the abundance-fitting loop and metallicity handling.

8. Run the test spectrum

Run:

```
./SPACE
```

After the run, check for output files:

```
ls -lh space*
```

Useful files include:

```
space_TGM_ABD.dat  
space_model.dat  
space_residuals.dat  
space_msg.txt
```

The main results are in:

```
space_TGM_ABD.dat
```

This file contains convergence information, radial velocity, FWHM, S/N, chi-square, T_{eff} , $\log g$, $[M/H]$, and abundance columns.

9. Interpret the convergence flag

The first column of the main output file is the convergence flag, usually called `conv`.

Use this rule:

```
conv = 0 or 1: result can be inspected  
conv > 1: fit failed or is outside the usable parameter/library range
```

Even when `conv = 0` or `conv = 1`, students must inspect the model plot, residuals, and continuum behavior before trusting the result.

10. Run the NGC 6569 shell scripts

Once the single-star test works, run the batch scripts from the working directory.

Standard conservative run:

```
bash run_best_conservative_no_norm_all_aat.sh
```

Alpha/metals companion run:

```
bash run_alpha_metals_best_conservative_no_norm_all_aat.sh
```

These scripts loop over the `aat_*.txt` spectra, write a fresh `space.par` for each fiber, run SPACE, save the output files, and append results to a master table.

11. Student checklist before accepting a fit

For every star, check:

1. Did SPACE produce a result file?
2. Is conv equal to 0 or 1?
3. Is the fitted model visually reasonable?
4. Are the residuals acceptable?
5. Is the spectrum inside the reliable parameter range?
6. Is the S/N high enough for the abundance being used?
7. Does $[\text{Fe}/\text{H}]_{\text{SP}}$ agree reasonably with $[\text{Fe}/\text{H}]_{\text{DP}}$?
8. Does the SED fit broadly agree with the spectroscopic solution?

Do not accept a SP_Ace result only because the program produced a number. The fit must be visually and scientifically inspected.

NOTE: For our project, the key warning is that the current official SP_Ace v1.4 documentation emphasizes the newer GCOG v2.2 library and a 4800–6860 Å range, while our AAT workflow used the CaT-region windows around 8450–8745 Å. The original SP_Ace paper describes the GCOG approach and includes red/near-IR coverage, but students should preserve the exact working GCOGLibrarySPACE1/ used for the NGC 6569 run unless they are deliberately testing a new library version.